

Exploring Machine Learning for Credit Card Fraud Detection from a Philippine Perspective

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Abstract

This study examines how machine learning (ML) techniques are applied in the Philippine setting to identify credit card fraud. This research aims to provide insights into the effectiveness of ML techniques in fraud detection, focusing on customizing ML algorithms to the distinct patterns and dynamics of credit card fraud in the Philippines, considering the nation's unique economic, technological, and social milieu. The research assesses the efficacy of different machine learning (ML) models using available data on credit card fraud occurrences and suggests improving fraud detection systems in Philippine financial institutions through ML integration. It also examines the opportunities and difficulties of using ML-driven fraud detection techniques in the Philippine financial industry.

CCS Concepts

• Security and privacy; • Human and societal aspects of security and privacy; • Privacy protections;

Keywords

Credit Card, Fraud Detection, Machine Learning, Philippines, Support Vector Machines, Artificial Neural Network, Prevention, Detection

ACM Reference Format:

Eric Blancaflor, Keziah Dawn Asuncion, Hailie Jade Reyes, and Michaela Verzosa. 2024. Exploring Machine Learning for Credit Card Fraud Detection from a Philippine Perspective. In *2024 The 6th World Symposium on Software Engineering (WSSE) (WSSE 2024), September 13–15, 2024, Kyoto, Japan*. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3698062.3698088>

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WSSE 2024, September 13–15, 2024, Kyoto, Japan

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ACM ISBN 979-8-4007-1708-6/24/09

<https://doi.org/10.1145/3698062.3698088>

1 Introduction

1.1 Background of the Study

Parallel to the phenomenal growth of digital transactions and on-line commerce, credit card fraud in the Philippines has risen to incredible heights. Together with increasingly sophisticated forms of fraudulent activities come growing requirements for equally innovative solutions that can cut these crimes down. In line with this, this research will investigate whether machine learning could be a powerful tool in enhancing credit card fraud detection within the Philippine financial sector. It is an artificial intelligence domain that imparted computers with the ability to learn from data without explicit programming, hence a good solution for identifying fraudulent patterns and anomalies in credit card transactions. Focusing on customizing the ML algorithms for the peculiarities of the Philippine context—economic, technological, and social—this paper, research will try to enhance the effectiveness of fraud detection systems. The literature review will go into the background of credit card fraud in this modern digital world, the scope and purpose of the research, the methodology, key findings and discussions, and finally, implications and future research directions. Credit card fraud instances are increasing at a rapid rate in the Philippines, which goes in direct relation to the rapid growth of its digital economy [1]. This threat increases with the limited effectiveness of traditional fraud detection. Herein lies the relevance of this review. By providing way to the possibilities of ML for the Philippine setting, this study hopes to add to the existing discussion regarding the strengthening of financial institutions from fraud in developing economies. Furthermore, lessons learned in tailoring ML models for the distinctly Philippine environment will also serve other developing countries that face similar challenges.

1.2 Objective of the Study

This paper identifies the escalating threat of credit card fraud in the Philippines and explores its parallelism with the proliferation of computer-based transactions in the country and simultaneously with the inadequacy of traditional fraud detection methods. It points out the main research question of whether it would be possible to fine-tune machine learning algorithms to the Philippine context for more effective fraud detection. This section reviews the background of the study on credit card fraud, its scope and purpose, methodology used, findings, and discussions. The paper seeks to

prepare a better ground for an all-rounded argument about ML's potential in fighting credit card fraud in the Philippine financial sector.

2 LITERATURE REVIEW

With the sudden surge of digital payment platforms, fraudulent credit card transactions in the Philippines have risen by 21% since the outbreak of COVID-19. Many of the fraud prevention systems do not consider the security of e-commerce networks and the fraud detection systems face issues like enormous quantities of data, skewed distribution, concept drift, and real-time detection assistance. Although at the cost of explainability, the accuracy achieved by Machine learning models, empowered with superior pattern detection and scalability, make them the future of Fraud detection. Future research shall investigate the impacts of additional sample techniques meta-classifiers and meta-learning methodologies to deal with highly imbalanced credit card fraud data.

2.1 Fraud Issues on Credit Card Usage

According to Philstar [2], MANILA, Philippines – Fraudulent credit card operations in the Philippines through remote and other digital payment channels increased by 21% since the start of the pandemic, according to the Credit Card Association of the Philippines. "The industry has suffered financially due to an increase in fraud cases brought on by the increasing acceptance of different digital payment platforms," Ilagan stated. According to ABS-CBN News [3], The Bangko Sentral ng Pilipinas said on Monday that online fraud cost consumers more than P540 million in 2021 alone. It also appealed to lawmakers to pass a bill that protects financial customers. BSP said there were 42,456 complaints from fraud victims of online monetary transactions between 2020 and 2021 involving phishing, identity theft, and malware attacks, among other forms of deceit. Most of these have already been considered closed. In addition, Diokno enlightened the senators that a lot of complaints had resolutions that did not benefit the consumer. To this extent, doubtless, the increase in digital transactions has made consumers' lives easier and faster, but this trend has also opened up room for new types of financial fraud. According to the CCAP, there has been a phenomenal surge in fraud-related activities, in particular through digital and remote payment channels [4]. Apart from being a threat to the consumers, the rise of these fraudulent cases is costing the sector financially.

Responding to this escalating threat, the BSP has started to address all of these concerns. With damages estimated at P540 million last year, cybercrime highlights the necessity of better policies for securing clients [3]. The law was legislated in response to the BSP's appeal to Congress to pass this type of law [5]. It provides for the rights of financial consumers for fair treatment and timely settlement of their complaints.

It is currently investigating claims of customers of Philippine bank BDO that they lost money to internet scammers who used accounts at Union Bank of the Philippines. It reported having monitored the increased complaints in the past week and subsequently frozen several accounts. The Philippines Anti-Money Laundering Council announced that the country had stepped up its fight against money laundering and terror financing in November 2021

[6]. Former RCBC branch manager Maia Deguito was in January 2019 convicted on eight counts of money laundering in the \$81 million theft from the Central Bank of Bangladesh. The bank was fined PHP 1 billion (\$19.17 million) in August 2016 for its failure to prevent the flow of stolen funds through it [6].

Maria Nite, president of Bank Capital Development Corp., was behind a notorious Treasury bill scam in which she bilked banks and investors by selling the Treasury bills repeatedly to trick people into believing they were all of the same T-bill. She made off with P2.5 billion (\$200 million) and was arrested by FBI agents in 1997, and extradited to the Philippines [6]. The FrancSwiss & Deutschers scam brought losses of PHP 1 billion, while that of the 12DP scam was \$500 million. Legacy insurance and pension scam caused losses of PHP 30 billion, or \$591 million. Aman Futures Group conned more than 15,000 investors in Mindanao and Visayas into a pyramid scheme where an investment return of 80% in just 20 days was promised for commodities trading. Connected to Legal Funds Transfer was the case of Annabella C. Ylagan, convicted in April 2018 of 55 counts of money laundering [6].

Between 2020 and 2021, card fraud losses globally increased by more than 10 percent. This is the highest rise since 2018. It is estimated that 30 billion dollars were lost by merchants and card acquirers; of those, losses of about 12 billion were from merchants and card acquirers in the United States alone, as reported. Please note that the figures shown here include both credit card fraud and debit card fraud. They are not distinguished on the website, and neither is there any statistic on the US, which is yet another country known for its addiction to credit cards [7].

From \$28.4 billion in 2020 and \$27.9 billion in 2018, a total of \$33.5 billion was reported lost due to card theft globally in 2022. In the EU, card transactions of € 1.53 billion were reported as fraudulent in 2021, representing a 7% decline in fraudulent card transactions overall. 93% of fraudulent charges in 2024 included credit cards issued in the United States that were still in the owners' possession. Card-not-present is the fraud type affecting \$174 million in lost revenue in 2023. In 2023, customers did report 368,379 cases of fraud related to online purchasing for a cost of \$392 million. 94% of all card-present transactions globally were made using cards with EMV chips as of the second quarter of 2023, compared to 92% at the same time in 2022 [8].

Despite such efforts, these are *prima facie* reasons to believe that more needs to be done to educate and create awareness among customers concerning the dangers associated with the use of e-financial services. Several resolved cases without victims receiving adequate compensation is a pointer to the need for more effective enforcement and consumer education campaigns. It puts customers on notice to be vigilant and at all times keep constant checks on their accounts and report suspicious activities to their banks or appropriate authorities. To have the capacity for a safe and reliable digital ecosystem, the collaboration of all concerned stakeholders—from financial institutions to regulators and consumers—should be assurance-based once digital finance makes its way to the country. Other than individual customer protection, a collaborative fight against fraud would further cement the integrity of the Philippine financial system.

2.2 Fraud Detection Studies on Machine Learning

Fraudsters frequently target e-commerce systems and use a range of underhanded tactics to get through. Using Fraud Prevention Systems alone is not sufficient for securing the network. Two possible combinations of FPSs and FDSs are sharing their capacities in securing e-commerce systems. However, the following factors usually make FDSs weak. Some of these are concept drift, the need for real-time detection, and imbalanced distribution, coupled with handling vast volumes of data. Furthermore, three specific limitations prevent FDSs from becoming highly efficient: the problems associated with the misclassification costs, noisy data, and overlapped classes. These constraints drive the models constructed improperly and to become too complicated detection models - affecting the predicted accuracy adversely. To address these problems, the following fraud detection tools are more commonly relied on. These tools are logistic regression, rule-induction techniques, decision trees, artificial neural networks, support vector machines, and meta-heuristic such as evolutionary algorithms. According to Abdallah et al. in 2016, the instruments are applied to most of the fraud detection industries [9]. Techniques provide a feasible means to improve the effectiveness of the FDSs in securing e-commerce platforms.

- The matthews correlation coefficient (mcc) is a metric to assess how well binary classifications are done. It is well recognized as a balanced measure that can be applied even in cases where the classes have drastically different sizes because it accounts for both true and false positives and negatives¹. This formula is used to calculate the mcc: $MCC = (TP \times TN) - (FP \times FN) / \sqrt{((TP + FP)(TP + FN)(TN + FP)(TN + FN))}$: where tp is true positives, tn is true negatives, fp is false positives, and fn is false negatives[10].
- Balanced classification rate (bcr): bcr is the average of the recall scores for each class. It is sometimes referred to as the average recall or the balanced accuracy. It helps handle datasets that are unbalanced³. It comes out to be: $BCR = \frac{2 \times \text{Sensitivity} + \text{Specificity}}{3}$. The genuine positive rate is called sensitivity, and the true negative rate is called specificity[11].
- A statistical model used in machine learning for binary classification issues is called logistic regression. Logistic regression forecasts an event's probability, which is a number between 0 and 1, as opposed to linear regression, which predicts a continuous outcome [12].
- K-nearest neighbor: k-nn is a supervised, non-parametric learning classifier that classifies or predicts how to categorize a single data point based on its proximity to other data points⁶⁷. When given a test observation and a positive integer k, k-nn finds the k nearest points and can conclude these closest targets [13].
- Using the bayes theorem as its foundation, naïve bayes is a probabilistic machine learning technique that may be used for various classification tasks. It is referred to as "naïve" because it assumes that each characteristic in the model is independent of the others [14].

Although the tradeoff between explainability and accuracy exists, with improved pattern detection and scalability, machine learning models remain the way forward for fraud detection. This wave of

machine learning-driven focus shift from pre-transaction to post-transaction detection portrays success in this war against fraud and creates a much safer digital transaction environment [15]. In this experiment, the three methods will be applied to both preprocessed and raw data. The implementation of the work will be done in Python. Their performance will then be evaluated using the approaches based on their accuracy, sensitivity, specificity, precision, Matthews Correlation Coefficient, and balanced classification rate. Descriptive statistics of models applied for logistic regression and k-nearest neighbor classifiers have their best accuracy rates at 54.86% and 97.69%, respectively, while that of naïve Bayes is at 97.92%. Comparative results show that the K-nearest neighbor outperforms the logistic regression and naïve Bayes methods. The research gives the effect of hybrid sampling on the binary classification of imbalanced data. Expected future areas of studies on meta-classifiers and meta-learning strategies for the handling of highly imbalanced credit card fraud data and, the effects of other re-sampling techniques could also be explored [16].

2.3 Focus on Fraud Prevention and Detection

Concerning the magnitude of the problem of credit card fraud in the Philippines, the strategic approach has been about strengthening fraud prevention and detection systems. It is through a combination of multifactor authentication, encryption, and real-time monitoring that FPSs become better equipped to detect, at an incredibly fast pace, any suspicious action. Advanced analytics, along with machine learning algorithms like Artificial Neural Networks (ANN) and Support Vector Machines (SVM), strengthens fraud detection capabilities within a Fraud Detection System (FDSs). These algorithms learn to recognize fraudulent patterns and anomalies in transaction data. Support Vector Machine is one of the popular methods that is used to recognize patterns and classify problems, especially when binary categorization like classifying fraudulent and nonfraudulent patterns is involved.

Like all other AI techniques, an SVM needs to be trained to return a model capable of predicting. Domains of application include but are not limited to the following: text categorization, bioinformatics, and face detection. SVM borrows from non-parametric applied statistics, neural networks, and the basics of machine learning. Its decision function, based on the theory of Structural Risk Minimization, will define a boundary between classes in the case of binary classification. This function also plays a very important role in determining the separation margin between classes, wherein it is learned by parameters w and b during the training phase by maximizing the margin [17]. Here is the sample of the SVM Equation: $f(x) = sqn(x.w) + b$

To scale these systems up to the next level, there is an urgent need to perform periodic upgrades of security protocols and algorithms to stay ahead of the tactics of fraudsters in vogue each day [18]. Big data technology can provide the competence of these systems to effectively deal with voluminous data on transactions, thus identifying fraud activities with better accuracy in real-time. This is because good collaboration within the bodies of banks, payment processors, and regulatory authorities is especially crucial in sharing vital information and trends on fraud. By doing so, it becomes possible to foster universally accepted standards and solutions that

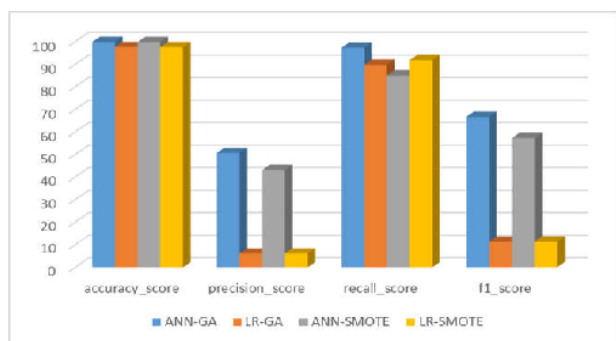


Figure 1: Comparative Performance of Machine Learning Models in Credit Card Fraud Detection

effectively mitigate any vulnerabilities across all the platforms. Training employees in financial institutions on the knowledge of current fraud detection technologies and techniques helps maintain an effective level of alertness and competence. Equally important are consumer education programs that can boost knowledge of security procedures and the importance of protecting personal information. Lastly, the development of machine learning adaptive models into applications able easily to acquire information on new fraudulent actions and, in the process, alter their algorithm will in the long term ensure very effective fraud detection systems [19].

2.4 Approach to Address Credit Card Fraud

Identification of complex challenges in mitigating credit card fraud—that this entails false alert cases and needs to be in real-time—is important. To overcome such challenges, developing and putting in place robust solutions will be relevant. These include real-time monitoring options that facilitate immediate identification of suspicious trends and reactions to the same. Furthermore, the improvement of rules in fraud prevention itself increases accuracy and makes the measures more responsive. These preventive steps can then be implemented to safeguard the business from fraud, fortify the protection of its financial assets, and uphold customer trust [20].

Figure 1 compares the performance of two Machine Learning models: Artificial Neural Network (ANN) and Logistic Regression (LR), with Genetic Algorithm (GA) and the Synthetic Minority Over-sampling Technique (SMOTE)-based enhancements to each model [21]. These enhancements are added to each model to make the model more accurate and robust to skewness—a relevant factor when dealing with fraud detection. It has the best performance compared to other models on all aspects of accuracy, precision, recall, and F1 score. This will mean it can detect fraud more impressively and with few misclassifications. This means that probably, ANN-SMOTE could probably be highly efficient, for example, in credit card fraud detection, where illegal transactions are very few as compared to legal ones. On the other hand, even if the LR models—both LR-GA and LR-SMOTE—exhibit relatively poor performance considering these metrics, they can turn out to be useful in applications for which the interpretability of the decisions made by a model is paramount. In particular, in regulatory compliance,

this may be as important as explaining the rationale behind decisions as accuracy. Therefore, in the case of credit card fraud, it would be wiser to rely more on models like ANN-SMOTE to enhance the possibilities of detection. However, including LR models is beneficial in meeting compliance and transparency requirements. Therefore, a middle path can be adopted so that the discoverability and prevention of fraud are more potent, and at the same time, the developed processes remain compliant with the set regulations.

2.5 Using Machine Learning to Avoid Credit Card Fraud

The case of credit card fraud detection in the Philippines stays timely due to the steep rise in fraudulent transactions by 21% since the pandemic [2]. Traditional FPSs do not suffice to protect digital payment networks; thus, there is a need for modern technologies like ML. The discussion points out the seriousness of the issue of fraud and the need for proactive actions based on insights from reliable sources—the Credit Card Association of the Philippines and the Bangko Sentral ng Pilipinas. Very promising results in enhancing fraud detection skills, despite inherent model explainability–accuracy trade-offs, have been seen lately with machine learning models, especially support vector machines and artificial neural networks. The discussion needs further research in more creative methods such as meta-classifiers or meta-learning strategies to solve the problems arising from fraudulent credit card use in the Philippines. It also calls for stakeholder engagement in advanced analytics to set the foundation for a collective defense against fraudsters on digital payment networks. The discussion is very much in keeping with the research aims to investigate machine learning in credit card fraud detection in the Philippines and provides very valuable insights and ideas for future research in this field.

In addition, the implementation of machine learning algorithms into fraud detection systems increases the accuracy of the identification of fraudulent actions, providing real-time detection with the capability of adaptive learning [22]. Machine learning models adapt to newly developed fraud techniques by continuously analysing transaction data and patterns. This adaptive ability is quite important in fighting more complex fraud techniques, given the fast-moving nature of digital transactions. Furthermore, fraud prevention requires collaboration between different stakeholders, including financial institutions, regulatory authorities, and technology providers. Only through shared insights, best practices, and collaboration can stakeholders be better placed to improve fraud prevention measures within digital payment networks and reduce vulnerabilities. Advanced analytics also showcases a greater interest in the use of data-driven techniques to unearth quite hidden trends and anomalies that serve as leading indicators of fraudulent activities, and therefore secure and preserve the integrity of financial transactions in the Philippines.

The researchers have taken a detailed drive through various machine learning models in service for credit card fraud detection. This section will mainly focus on how well these models perform in terms of accuracy—one of the major factors in determining how efficient each model is at the detection of fraudulent transactions. Further, some strategic ideas that are aimed at enhancing the forecast accuracy and total efficiency of such models are sought. The

Table 1: Evaluating Machine Learning Models for Credit Card Fraud Detection: Accuracy and Improvement Strategies

Model	Dataset Description	Method	Accuracy	What to Improve	Reference
Logistic Regression	Credit card transaction data from the Philippines	Supervised binary classification	54.86%	Explore hybrid sampling to handle imbalanced data	[23]
K-Nearest Neighbors (k-NN)	Data points clustered based on proximity to detect fraud	Non-parametric, nearest neighbor	97.69%	Increase the diversity and size of the dataset for better generalization	[24]
Naïve Bayes	Transaction patterns focusing on anomaly detection	Probabilistic classification	97.92%	Adjust independence assumptions in the model	[25]
Support Vector Machines (SVM)	Features intended to distinguish between authentic and fraudulent transactions	Kernel-based learning technique	Varies up to 80%	Incorporate meta-learning for skewed data handling	[26]
Artificial Neural Networks (ANN)	High-dimensional datasets for pattern recognition	Deep learning algorithms	Varies up to 98.44%	Optimize neural network architecture for real-time detection	[27]

authors have clearly shown that, for instance, some of their models address imbalanced datasets or improving algorithm parameters as the critical areas to enhance, and therefore this gives a clear avenue for future research on its practical application in fraud detection. This thorough assessment improves our understanding of the current capabilities and limitations of existing machine learning models and further drives innovation in methods to secure financial security systems against the threats of fraudulent activities.

3 CONCLUSION

In conclusion, the digital payment systems that have been invented have proved to be a potential source of unintended consequences for increased fraudulent activities, especially in the Philippines, which, since the epidemic, has seen an increase in credit card fraud by 21%. Current FPSS are proving to be utterly inadequate in their security for e-commerce networks, while various challenges, such as managing huge amounts of data, skewed distribution, concept drift, and the demand for real-time detection, plague FDSS. However, the introduction of machine learning models puts the future of fraud detection on the bright side. The application of models that can learn from data without explicit programming for improving security in e-commerce platforms is increasingly used. In this way, an advanced war against fraud forms a safer environment for digital transactions with a transition from pre-transaction to post-transaction detection. This study, therefore, is much needed in the Philippines, which is classified as one of the countries with a large volume of fraudulent credit card transactions. In essence, what this means is that there should be an enhanced FDSS with advanced analytics, machine learning algorithms, and techniques for anomaly detection while strengthening the FPSS by multifactor authentication, encryption protocols, and real-time monitoring. This would involve collaboration concerning insight, best practices, and threat information sharing among financial institutions, regulatory bodies, and technology providers. The research goes further to outline how machine learning will benefit fraud detection in the financial sector. Tailoring machine learning algorithms to the intricacies of the Philippine context—the unique economic, technological, and social aspects—is what can greatly increase the efficacy of fraud

detection systems. Results from this study may help related issues dealing with other developing countries.

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